

L2 – Trig Ratios for Angles Greater than 90°

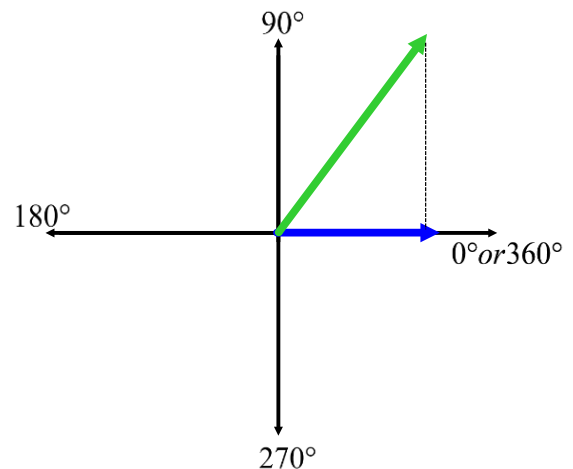
MCR3U

Part 1: Reference Angles

Initial Arm: Always lies on the positive x -axis at 0° .
Meets the terminal arm at the origin.

Terminal Arm: The arm that rotates around the origin counter clockwise to form a positive angle or clockwise for a negative angle.

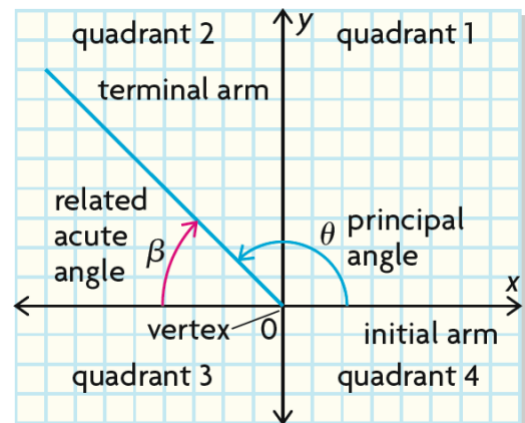
Angle θ is measured from the initial arm to the terminal arm.



Principal Angle: The counter clockwise angle between the initial arm and the terminal arm of an angle in standard position. It's value is between 0° and 360° .

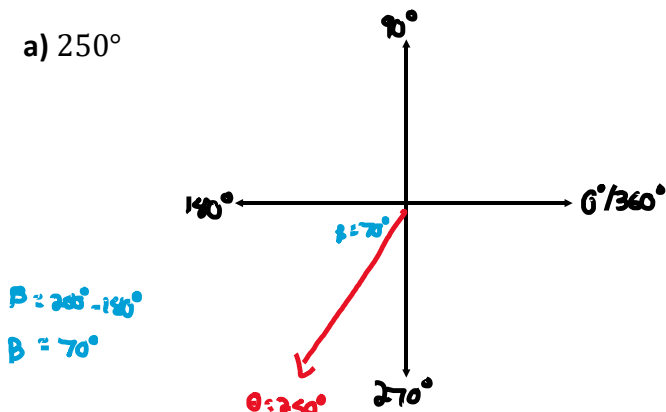
Related Acute Angle (reference angle): The acute angle between the terminal arm of an angle in standard position and the closest x -axis when the terminal arm lies in quadrant 2, 3, or 4.

The reference angle helps us determine the exact trig ratios when we are given obtuse angles.

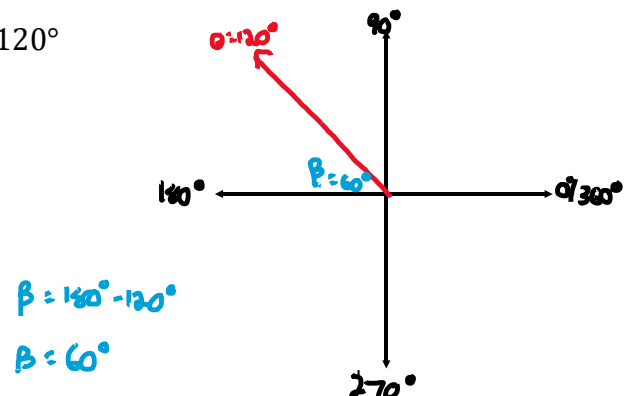


Example 1: Find the reference angle for each of the following principal angles

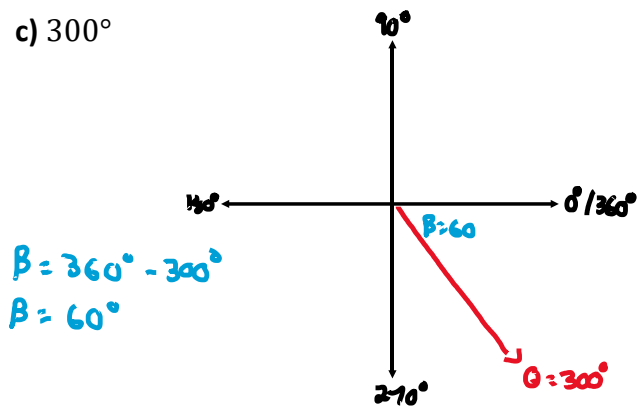
a) 250°



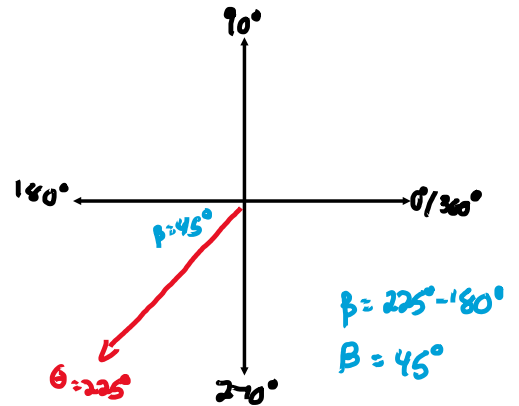
b) 120°



c) 300°



d) 225°



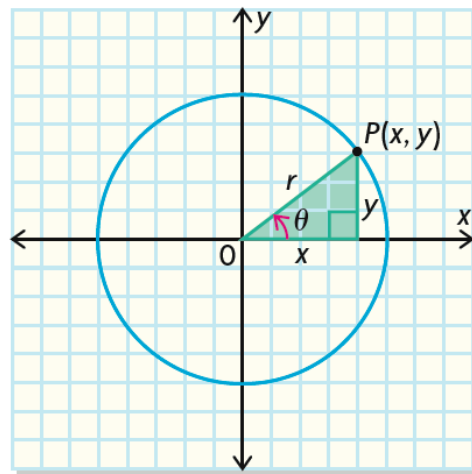
Part 2: Evaluating Trig Ratios for Any Angle

For any point $P(x, y)$ in the Cartesian plane, the trigonometric ratios for angles in standard position can be expressed in terms of x , y , and r .

$$\sin \theta = \frac{y}{r}$$

$$\cos \theta = \frac{x}{r}$$

$$\tan \theta = \frac{y}{x}$$



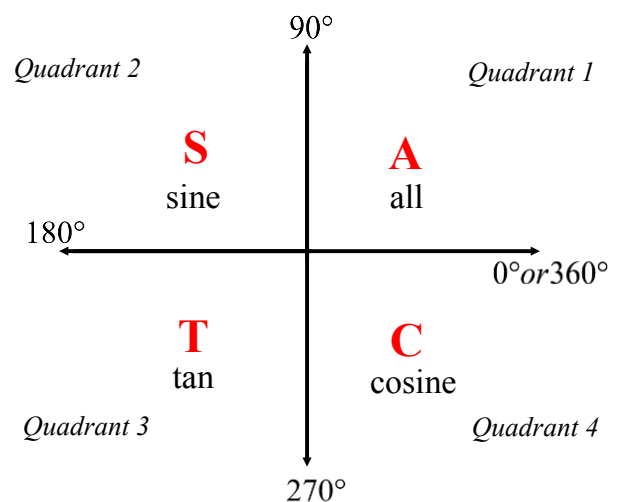
The CAST rule is an easy way to remember which primary trig ratios are positive in which quadrant. Since r is always positive, the sign of each primary ratio depends on the signs of the coordinates of the point (x, y) .

In Q1, **All** ratios are positive because both x and y are positive.

In Q2, only **Sine** is positive, since x is negative and y is positive.

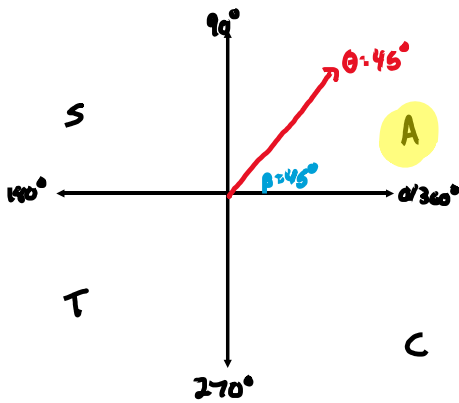
In Q3, only **Tangent** is positive, since both x and y are negative.

In Q4, only **Cosine** is positive, since x is positive but y is negative.



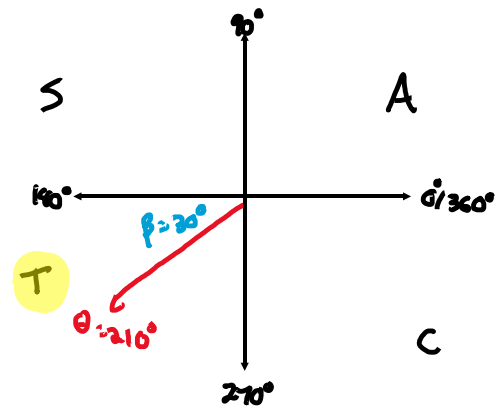
Example 2: Find the EXACT value of each of the following

a) $\sin 45^\circ$



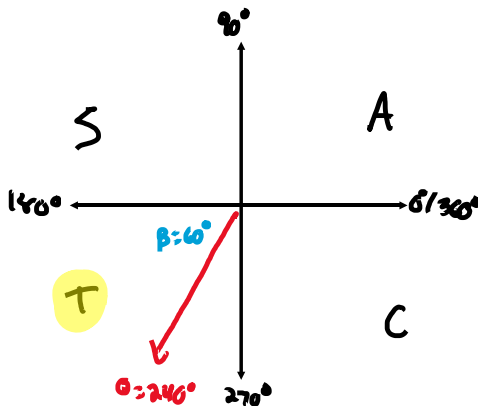
$$\sin 45 = \frac{1}{\sqrt{2}}$$

b) $\sin 210^\circ$



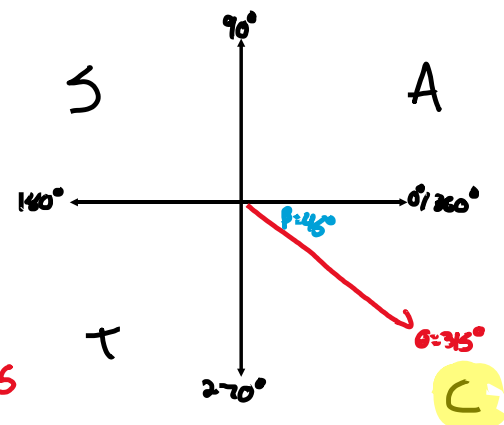
$$\begin{aligned}\sin 210 &= -\sin 30 \\ &= -\frac{1}{2}\end{aligned}$$

c) $\cos 240^\circ$



$$\begin{aligned}\cos 240 &= -\cos 60 \\ &= -\frac{1}{2}\end{aligned}$$

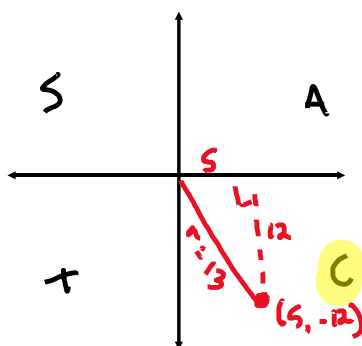
d) $\tan 315^\circ$



$$\begin{aligned}\tan 315 &= -\tan 45 \\ &= -1\end{aligned}$$

Example 3: Each point lies on the terminal arm of angle θ in standard position. Determine each of the primary trig ratios for angle θ .

a) $(5, -12)$



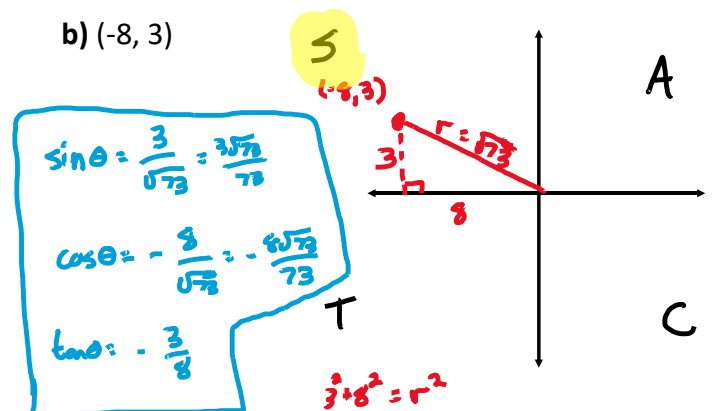
$$\begin{aligned}5^2 + 12^2 &= r^2 \\ 169 &= r^2 \\ r &= 13\end{aligned}$$

$$\sin \theta = -\frac{12}{13}$$

$$\cos \theta = \frac{5}{13}$$

$$\tan \theta = -\frac{12}{5}$$

b) $(-8, 3)$



$$\sin \theta = \frac{3}{\sqrt{73}} = \frac{3\sqrt{73}}{73}$$

$$\cos \theta = -\frac{8}{\sqrt{73}} = -\frac{8\sqrt{73}}{73}$$

$$\tan \theta = -\frac{3}{8}$$

$$\begin{aligned}3^2 + 8^2 &= r^2 \\ 73 &= r^2 \\ r &= \sqrt{73}\end{aligned}$$

Part 3: Unit Circle

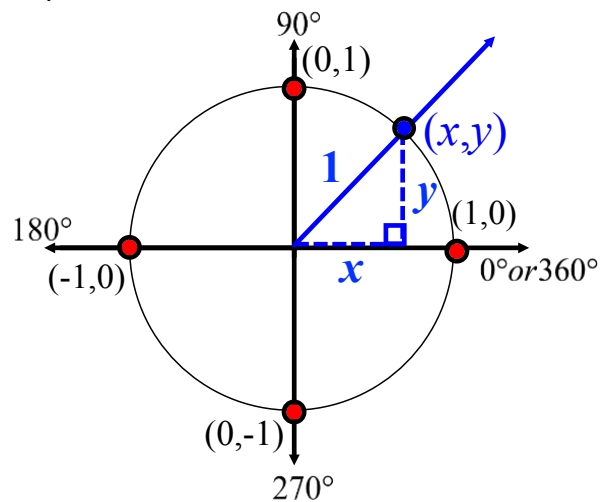
The unit circle, a circle with a radius of **1 unit**, is very useful since the x and y coordinates of where the terminal intersects it tell us the Cosine and Sine ratios respectively.

For any point $P(x, y)$ in the Cartesian plane that intersects the **unit circle**, the trigonometric ratios for angles can be expressed in terms of x , y , and r .

$$\sin \theta = \frac{y}{1} = y$$

$$\cos \theta = \frac{x}{1} = x$$

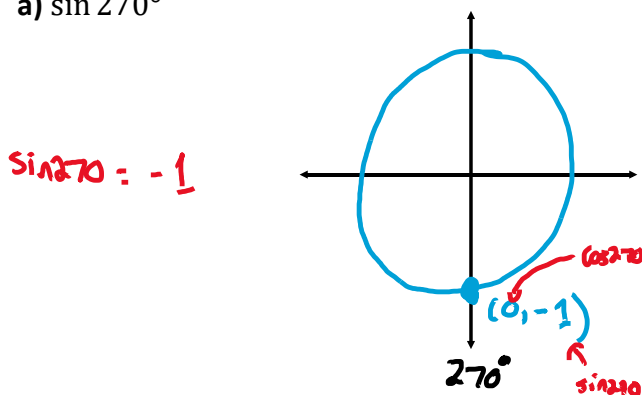
$$\tan \theta = \frac{y}{x}$$



<http://www.mathsisfun.com/geometry/unit-circle.html>

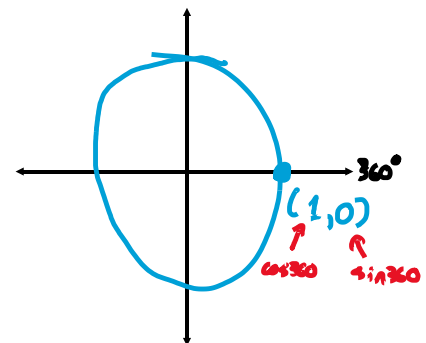
Example 4: Find the EXACT value of each of the following

a) $\sin 270^\circ$



b) $\cos 360^\circ$

$$\cos 360 = 1$$



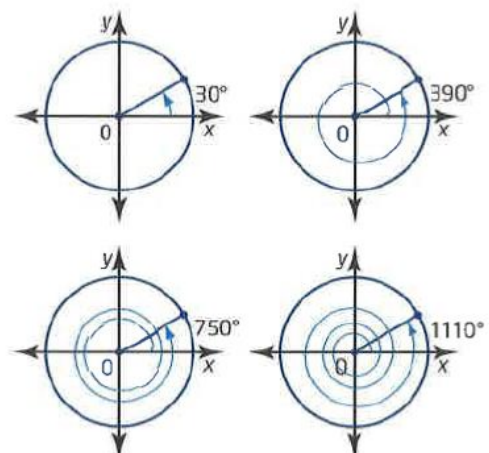
Part 4: Negative and Co-terminal Angles

Co-terminal angles are angles in standard position that have the **same terminal arm**.

Starting at 30° and rotating 360° counter clockwise will bring you back to the same terminal arm.

$$30^\circ + 360^\circ = 390^\circ$$

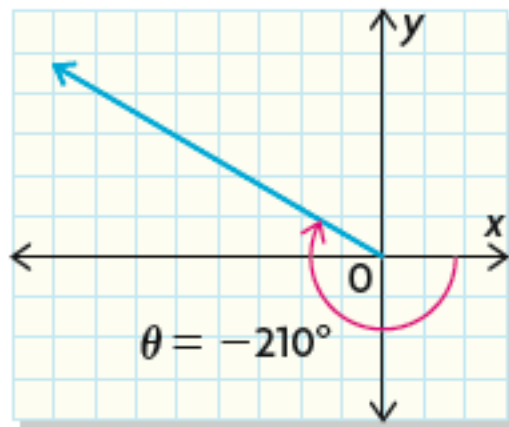
Therefore, 30° and 390° are co-terminal.



A negative angle is an angle measured clockwise from the positive x -axis.

You can find an equivalent (co-terminal) positive angle by adding 360° to the negative angle.

-210° and 150° have the same terminal arm (co-terminal) and therefore have the same trigonometric ratios.



Example 5: Find three co-terminal angles of 60°

$$\theta_1 = 60^\circ + 360^\circ$$

$$\theta_1 = 420^\circ$$

$$\theta_2 = 60^\circ + 2(360^\circ)$$

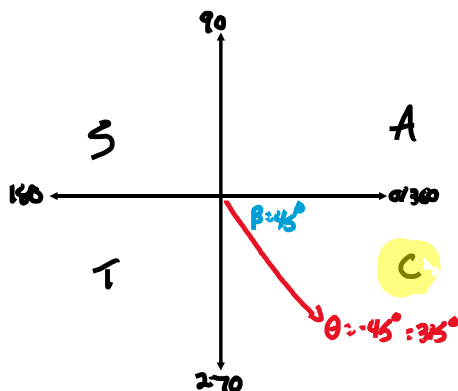
$$\theta_2 = 780^\circ$$

$$\theta_3 = 60^\circ + 3(360^\circ)$$

$$\theta_3 = 1140^\circ$$

Example 6: Find the EXACT value of each of the following

a) $\sin(-45^\circ)$



$$\sin(-45^\circ) = \sin(-45^\circ + 360^\circ)$$

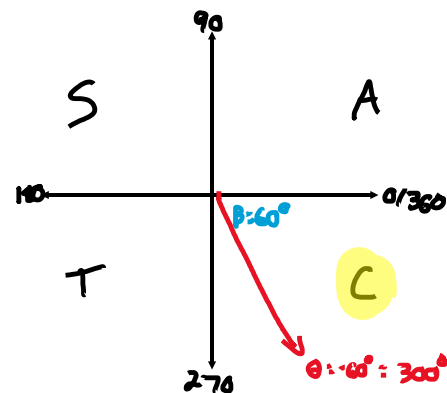
$$= \sin(315^\circ)$$

$$= -\sin(45^\circ)$$

$$= -\frac{1}{\sqrt{2}}$$

$$= -\frac{\sqrt{2}}{2}$$

b) $\cos(-60^\circ)$



$$\cos(-60^\circ) = \cos(-60^\circ + 360^\circ)$$

$$= \cos(300^\circ)$$

$$= \cos(60^\circ)$$

$$= \frac{1}{2}$$